

VIVEK VIJAYKUMAR INGLE

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SUMMARY

AI & Robotics Engineer specializing in Imitation Learning, ROS development, and full-stack AI systems. Proven track record of delivering production-ready robotic automation solutions and designing end-to-end ML pipelines. Experience spans edge AI deployment, computer vision, SLAM, and human-robot interaction systems for healthcare and industrial applications.

TECHNICAL SKILLS

Languages: Python, C++, C#, MATLAB, JavaScript, LaTeX

AI/ML: Imitation Learning, Computer Vision, TensorFlow, PyTorch, Model Optimization, Edge AI Deployment

Robotics: ROS/ROS2, SLAM, Motion Planning, Robot Control, Sensor Fusion, Pick-and-Place Systems

Development: Full-Stack Development, Streamlit, OPC UA, REST APIs, Git, CI/CD, Docker

Tools & Platforms: Unity, VR/AR Development, Google Cloud AI, Embedded Systems, Real-time Systems

PROFESSIONAL EXPERIENCE

MYWAI, Sestri Levante, Italy

AI & Robotics Engineer

November 2024 – Present

- Design and implement advanced AI algorithms for industrial robotic automation, specializing in Imitation Learning techniques for pick-and-place applications with 95%+ task success rates
- Build end-to-end ML pipelines from data collection through model training to deployment on edge devices, optimizing inference for real-time robotic control with <50ms latency
- Integrate pre-trained models and leverage APIs to accelerate development cycles by 40%, delivering scalable robotic solutions for Europe's fastest-growing Generative & Edge AI platform
- Lead technical architecture decisions for AI-driven automation systems serving multiple industrial clients

Robotics Software Engineer (Internship)

May 2024 – November 2024

- Developed and optimized ROS2-based robotic control systems supporting advanced Imitation Learning workflows, reducing development iteration time by 60%
- Designed and implemented full-stack Streamlit-based Imitation Learning toolkit enabling streamlined experimentation and data visualization for robotics researchers
- Successfully delivered MVP earning transition to full-time role; toolkit now used across organization for rapid prototyping and model evaluation

Heidelberg University, Heidelberg, Germany

Erasmus+ Research Trainee (Master's Thesis)

October 2022 – March 2023

- Designed and implemented haptic immersive VR scenarios with bimanual robotic exoskeleton for stroke rehabilitation under Prof. Lorenzo Masia at ARIES Lab
- Developed robotic mirror therapy system combining VR exergames with real-time haptic feedback, improving patient engagement scores by 35%
- Programmed complex control algorithms for multi-DOF exoskeleton systems ensuring safe, adaptive human-robot interaction for clinical applications

KEY PROJECTS

Monocular SLAM with TensorFlow Object Detection | Python, C++, TensorFlow, ORB-SLAM2, OpenCV

- Engineered portable 3D environment mapping system combining monocular SLAM (ORB-SLAM2) with real-time object detection using TensorFlow, utilizing only standard smartphone camera to eliminate need for specialized depth sensors or LiDAR
- Achieved real-time performance with <30ms processing time per frame, enabling low-cost spatial mapping and object recognition suitable for resource-constrained mobile robotics and AR applications
- Implemented robust feature tracking and loop closure detection algorithms, improving mapping accuracy by 25% in challenging lighting conditions and dynamic environments
- Validated system performance through comprehensive real-world testing scenarios including indoor navigation, object localization accuracy measurements, and multi-session map consistency evaluation

Human Activity Recognition for Activities of Daily Living (ADL) | Python, Scikit-learn, OpenCV, Linux

- Developed comprehensive Human Activity Recognition system using sensor fusion (accelerometer, gyroscope) and video data to identify Activities of Daily Living including walking, sitting, standing, and brushing teeth for assistive robotics
- Implemented complete machine learning pipeline with feature engineering, model selection, hyperparameter tuning, and cross-validation, achieving 92% classification accuracy across 12 activity classes
- Designed real-time processing architecture capable of processing sensor streams at 50Hz while maintaining low latency (<100ms) for immediate activity classification and response
- Enhanced capabilities of assistive robotic systems for elderly and special-needs care through intelligent activity detection, enabling proactive assistance and safety monitoring

EDUCATION

Master's Degree in Robotics Engineering

September 2019 – July 2024

Università degli Studi di Genova, Genoa, Italy

Erasmus+ Exchange: Heidelberg University, Germany (October 2022 – March 2023)

Bachelor of Technology in Electronics & Media Technology

2012 – 2016

Karunya Institute of Technology and Sciences, India

Diploma in Game and Interactive Media Design

2017 – 2018

FX School

CERTIFICATIONS & ACHIEVEMENTS

- **Erasmus+ Scholar:** Selected for competitive robotics engineering exchange program at Heidelberg University
- **Google Cloud:** Innovating with Google Cloud AI Certification
- **Industry Experience:** Electronic Arts Software Engineering Virtual Experience Program
- **Additional Certifications:** C# & Unity Games Engine, Exploratory Data Analysis with MATLAB